

Exotic pests of timber include termites and insect borers such as beetles and wasps. These pests can arrive in timber and timber products as eggs or larvae. These pests may go unnoticed until frass (a sawdust-like substance produced by feeding wood boring insects) and holes appear or chewing sounds are heard.

It is critical to manage the risk of these pests. These pests may be harmful to our environment and economy if they enter Australia, spread, and become established.

The import conditions published in our [Biosecurity Import Conditions \(BICON\)](#) system list approved treatment options for timber. Treatment options can vary according to the timber product, species and source country. Import conditions will outline the treatment rate/dosage and treatment provider requirements.

Approved treatment options

The following treatments are approved for treating timber and timber products as specified in [BICON](#).

Methyl bromide is used as a pesticide to mitigate the risk of timber pests.

Fumigation is applied as a gas that surrounds and penetrates the goods. This occurs within a gas-tight fumigation enclosure to retain the fumigant for the duration of the exposure period and maintain fumigant concentration levels.

This treatment cannot be applied to timber products that are painted, varnished or glazed. These coatings inhibit the penetration of the fumigant.

Sulfuryl fluoride is used as a pesticide to mitigate the risk of timber pests.

Fumigant is applied as a gas within a gas-tight fumigation enclosure, like methyl bromide fumigation.

Treatment of timber must be performed using an [approved third party software](#) to achieve the required concentration-time product (CT).

This treatment cannot be applied to timber products that are painted, varnished or glazed. These coatings inhibit the penetration of the fumigant.

Ethylene oxide is used as a pesticide to mitigate the risk of timber pests and pathogens (e.g. fungi).

Fumigant is also applied as a gas within a gas-tight fumigation enclosure. The timber is placed in a chamber under a vacuum (all air removed) and gas is introduced at controlled rates.

This fumigant can be used on timber products that are painted, varnished or glazed. These coatings do not inhibit the penetration of the fumigant.

Dry heat treatment and kiln drying are used to mitigate the risk of timber pests and pathogens.

The process involves heating the timber to reach an approved core temperature. This occurs within a temperature-controlled chamber or oven, or shipping container in some cases.

Thickness of the timber can alter the duration of the treatment.

Gamma irradiation is an effective treatment for timber pests and pathogens.

Gamma irradiation is a form of electromagnetic radiation that breaks down cellular DNA. This inhibits cell division (pathogens) and reproduction in multicellular organisms (e.g. insects).

This treatment does not use chemicals or pressure. It does not change the temperature of the product and can be applied to goods that are packaged.

Permanent preservatives formulations are a combination of pesticides and fungicides. As such permanent preservatives are an effective treatment for timber pests and pathogens.

[Permanent timber preservatives](#) are applied to timber through pressurised impregnation or glue-line treatments.

This treatment can be performed by any commercial provider that meets the [requirements for permanent preservative treatment](#).

Treatment providers

Offshore treatments must be performed by an approved treatment provider, or a provider endorsed by the exporting country's National Plant Protection Organisation (NPPO). Check [BICON](#) for the treatment provider requirements for the chosen treatment.

Onshore treatments must be performed under an [approved arrangement](#). Biosecurity directions for treatment are issued by the department.

For further information

Read more about:

- [treating cargo outside of Australia](#)
- the [methodologies and documents for biosecurity treatments](#)
- the [documentary requirements for imported goods](#).

You can search for [offshore treatment providers](#) and [onshore treatment providers](#).

Source: <https://www.agriculture.gov.au/biosecurity-trade/import/goods/timber/approved-treatments-timber>